6. Database Design

If storage is required, a simple database can contain User, Preference and Plan tables. Each plan can have a creation date and generated content.

Tabulation

Stage	Action
Input	Collect information
Process	Validate and generate prompt
AI	Send request to Gemini
Output	Display plan

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
INPUT → PROCESS → AI → OUTPUT
```

7. Security Design

The Gemini API key must remain in an environment variable or secure server configuration. User input should be validated, and sensitive information should not be unnecessarily collected.

Tabulation

Output Section	Example
Schedule	Weekly activity arrangement
Activity	Selected general activities
Recovery	Rest and recovery suggestions
Reminder	General wellness guidance

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
WEEKLY PLAN → ACTIVITY → RECOVERY → WELLNESS
```

8. Example Technical Flow

HTML form → Flask route → validation → Gemini client → generate_content → response.text → HTML result page.

Tabulation

Area	Example
User	User preferences
System	Prompt + Gemini response
Output	Formatted plan
Future	History/reminders/dashboard

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
SECURE BACKEND → GEMINI → DATABASE/OUTPUT
```